Rudransh Agrawal

agraw115@purdue.edu | +1 832-748-6900 | https://www.linkedin.com/in/rudranshagrawal

Education

Purdue University – ***West Lafayette, IN***

Master of Science in Computer Engineering - May 2024 *GPA : 3.81/4.00*

- Graduate Teaching Assistant for Microprocessor Systems and Interfacing (ECE 36200)

Bachelor of Science in Computer Engineering - May 2023 *GPA : 3.91/4.00*

- Eli Shay and RCA Zworykin Scholar
- Resident Advisor
- Undergraduate Teaching Assistant for Electrical Engineering Fundamentals I (ECE 20001)

Industry Experience

Amazon Web Services (AWS) **July 2025 – Present**

Software Development Engineer, Hardware Engineering (Accelerated Platforms) *Cupertino, CA*

- Design and implement firmware for Baseboard Management Controllers (BMCs) on EC2 servers with AMD and Nvidia GPUs, enabling new product introduction (NPI) for accelerated compute platforms.
- Develop PCIe integration and custom telemetry systems for GPU health, asset monitoring, and automated vision reporting.
- Engineer advanced fan control algorithms (FSC) for energy-efficient server cooling under high-performance loads.
- Own BMC firmware rollout and validation from prototyping to fleet deployment for new EC2 instance types.
- Collaborate across hardware, software, and infrastructure teams to ensure secure, scalable GPU fleet management.
- Integrate real-time telemetry data pipelines and PCIe subsystem support for system reliability and remote diagnostics.
- Maintain and continuously improve BMC features critical to AWS accelerated server operations.

Milwaukee Electric Tool Corporation **August 2024 – June 2025**

Firmware Engineer I *Chicago, IL*

- Designed and deployed embedded firmware for Power Interface Boards (PIB) to manage battery packs and orchestrator modules.
- Implemented deterministic device control with custom task kernels for safety-critical environments.
- Integrated digital I/O, ADC, and serial drivers under strict timing and safety constraints.
- Developed C++ infrastructure for CAN command and telemetry (PDO model).
- Led design reviews and collaborated with hardware teams for rapid architecture iteration.
- Automated regression testing with Python-based Hardware-in-the-Loop (HIL) test suites to boost system reliability.

Milwaukee Electric Tool Corporation **June 2023 – August 2023**

Embedded Software Engineering Intern *Brookfield, WI*

- Built MicroPython-based runners and UART-linked C++ APIs for embedded device testing on Arduino Teensy 4.1.
- Debugged and improved firmware using C/C++ and Linux toolchain.
- Developed automated CI/CD test scripts with Azure DevOps, reducing regression cycles by 80%.
- Collaborated in agile teams to identify, fix, and validate firmware across multiple hardware platforms.

Coursework

Embedded Systems, Real-Time Operating Systems, Advanced C Programming, ASIC Digital Design Lab, Microprocessor Systems and Interfacing, Computer Architecture, Firmware Development, Hardware-in-the-Loop Testing, Object-Oriented Programming in C++, Software Engineering, Applied Algorithms

Skills

Languages: C | C++ | Python | Bash
Embedded Systems: Real-Time Systems | RTOS (FreeRTOS) | Interrupt-Driven Programming | Bare-Metal Firmware | Board Bring-up
Communication Protocols: CAN | UART | SPI | I2C | PWM | ADC/DAC
Operating Systems: Embedded Linux | Yocto | Buildroot
Tools & Debugging: Git | Oscilloscopes | Logic Analyzers | GDB | In-Circuit Debuggers
DevOps and CI/CD: Azure DevOps | Jenkins | Docker | Test Automation
Software Engineering Practices: Test-Driven Development | Version Control | Agile/Scrum